



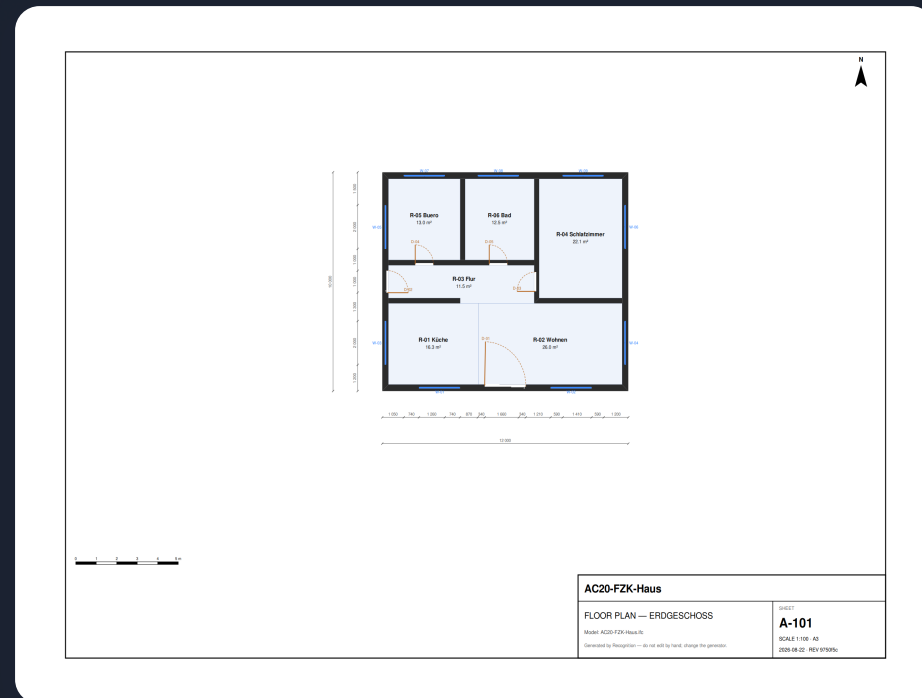
Recognition

your AI architect

Describe a building in plain words. Get four verified blueprints, a 3D model, and files a builder can open — with nobody in the loop between your sentence and the artifacts.

Devin plans it · deterministic code draws it · a cited rule engine verifies it

0-uddeshya-0.github.io/Recognition



A-101 · generated by the pipeline, no hand edits

Every industry has an engineering problem

SOFTWARE

has the loop

`write → run → test → fix`

Every attempt gets a verdict in seconds, so a wrong answer announces itself — which is exactly why an agent can work unsupervised.

ARCHITECTURE

had none

`draw → submit → wait → find out`

A layout that breaks building code surfaces at permit review or on site — months later. Small practices and their clients wait weeks for a first compliant answer, and every change restarts the wait.

Pick a domain with no way to check the work, and an agent is only guessing.

Building code is machine-checkable

Clear heights. Daylight ratios. Door widths. Corridor clearances. Which rooms a dwelling must contain. Every one of them is decidable from a model — an executable predicate, not an opinion.

The verdict already existed.
Nothing was sitting inside the loop.

Recognition is the layer that puts an agent inside it.

THE RULEPACK IN FORCE

BayBO Art. 45 (1)	law
clear height ≥ 2.40 m	
BayBO Art. 45 (2)	law
glazing ≥ 1/8 of floor	
BayBO Art. 46	law
kitchen + bathroom required	
BayBO Art. 48 (1)	law
> 2 homes → barrier-free	
DIN 18040-2	standard
1.20 m clear corridor	
Richtwerte	guidance
typical room areas	

Say it. Choose. Build.

1

Say it

“An office for my startup — 8 of us, an open studio, a meeting room, a small kitchen.” It asks only what it genuinely can't infer.

2

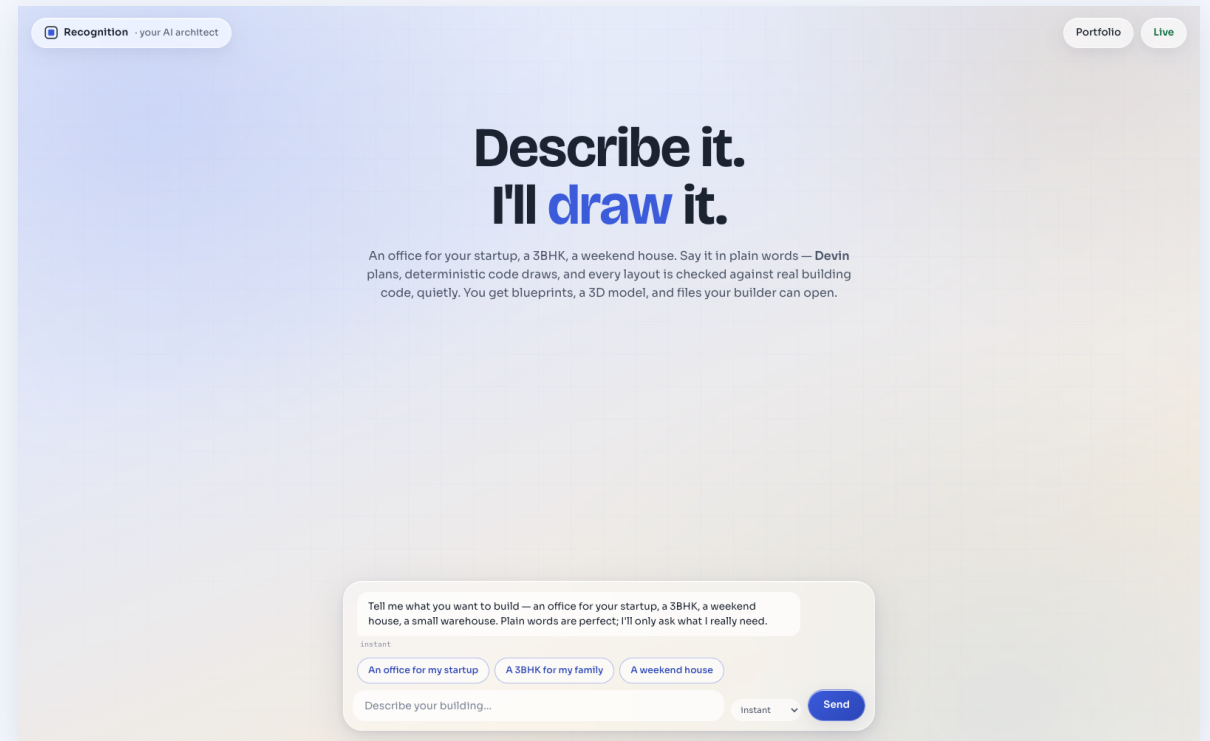
Choose

Four structurally different blueprints come back, dimensioned and already checked. Keep talking and it redrafts.

3

Build

Pick one: the 3D model appears with the plan below it. Archive it and the pipeline verifies, merges itself, and files it.



Live, in the browser — nothing to install, no account, no key.

Four takes — every one already checked

Recognition · your AI architect

PortfolioLive

4 takes on your brief



Compact core ✓ 57 checks · 2 unchecked

Compact core: Squarer envelope with a central hall spine. Least external wall per m2, so cheapest to build and to heat.

143 m² 8 rooms 75% usable

Build this one in 3D



Linear frontage ✓ 57 checks · 2 unchecked

Linear frontage: Wider, shallower envelope. Every habitable room reaches an exterior wall, which makes the daylight rule easy to satisfy.

144 m² 8 rooms 83% usable

Build this one in 3D



Open plan ✓ 54 checks · 2 unchecked

Open plan: No dedicated corridor; living spaces connect directly. Converts circulation area into usable floor.

147 m² 7 rooms 86% usable

Build this one in 3D



Generous plan ✓ 57 checks · 2 unchecked

Generous plan: Squarer envelope, every room a size up and a wide hall. Comfort over cost — the most building for the same programme.

175.5 m² 8 rooms 77% usable

Build this one in 3D

Tell me what you want to build — an office for your startup, a 3BHK, a weekend house, a small warehouse. Plain words are perfect; I'll only ask what I really need.

instant

This is a drafting round someone shared — 4 takes on their brief. Open one to see it in 3D, or describe your own building and I'll draft fresh ones.

instant

Describe your buildinginstantSend

Structurally different

Compact core, linear frontage, open plan, generous — different topologies, not the same plan ±10%.

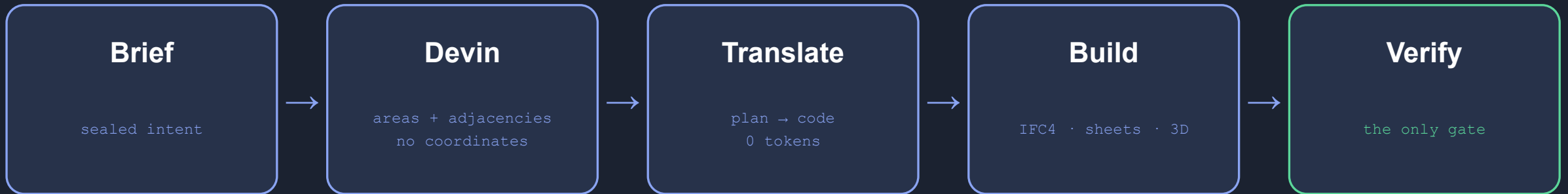
Quietly verified

Each card carries its own check count. Regulations stay underneath; the citations are one tap away.

Never a bare PASS

What the model can't see is counted and shown, never silently passed.

The model plans. The code draws.



A session cannot fake a pass.

Devin never emits a coordinate and never states a regulation — it returns a typed plan. The IFC model, the drawings and the verdict are produced afterwards by deterministic code it never ran, and judged against a cited ruleset it cannot edit. “I fixed it” is not a claim the system can accept from a model.

Four roles, one autonomous loop

Interviewer

Reads the client's own words and returns a typed brief — questions only where a rule genuinely needs one, every inferred value registered as an assumption.

Architect ×4

Four sessions in parallel, each handed a structurally different strategy. Each returns an ArchitectPlan: rooms, areas, adjacencies. No coordinates.

Critic

Reviews the winner for what a checker cannot see — a room you can't furnish, a route that doesn't work. Advisory by construction; it can annotate, never overturn.

Repairer

A contract error goes back verbatim as the repair prompt. Observed live: one rejection, one repair, clean pass — no human saw it.

Taught by the repo itself: four Agent Skills Devin discovers natively, playbooks per role, structured output enforced at the API, and Fusion-shaped delegation — judgement in the main agent, mechanical verification to the sidekick.

Trigger to merged artifact, nobody in between

3

Devin sessions
planned in parallel

2

rejected by the gate
— not by a person

1

pull request opened and
merged by github-actions

0

humans in the loop
after the trigger

THE RECORDED RUN · engine=devin · run 32600313272

One workflow_dispatch started three Devin sessions. Each returned a plan; deterministic code built and verified all three; the scorer picked the winner; the workflow opened a pull request and merged it — then deployed the result to the live site.

There is no Approve button anywhere in the product.

It knows good from bad — and what it cannot know

EVERY RULE CARRIES A TIER AND A CITATION

law a cited statute article

blocks a merge

standard a DIN norm, once triggered

blocks when in force

guidance published Richtwerte

never blocks

house our own convention

never blocks

A rule without a citation fails to load. No model may author one.

NO RUN EVER SHOWS A BARE PASS

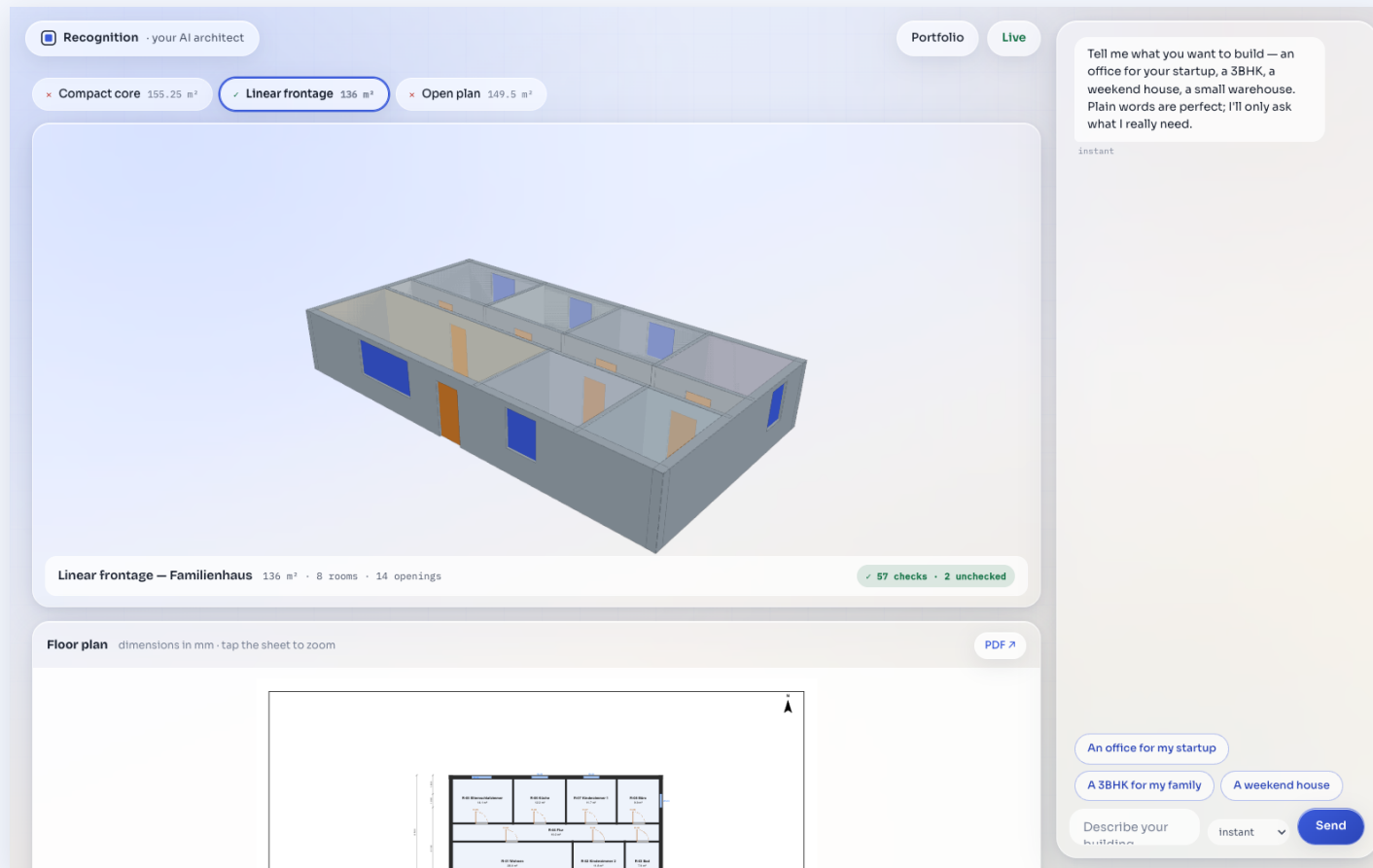
57 checked · 2 not evaluated · 0 failed

Smoke detectors and thresholds cannot be judged from a model, so they are reported — never counted as passes. The coverage travels with every verdict.

THE RULES CONFIGURE THEMSELVES

A brief with three homes auto-triggers BayBO Art. 48: DIN 18040-2 becomes mandatory, and the same 1.15 m corridor finding flips from advisory to blocking. Nobody configured that — the rule declares its own trigger.

Output an architect can actually use



`model.ifc`

IFC4 with real IfcSpace / IfcWall / IfcDoor — opens in Revit or ArchiCAD

`sheets .dxf`

AIA-named layers, millimetres — straight into AutoCAD

`sheets .pdf .svg`

dimensioned plans with a title block

`schedules .csv`

room, door and window schedules

`verdict.json`

every finding with its tier and statute citation

`design.py`

the building as readable, diffable code



Find an industry. Give it an engineer.

Architecture had the loop all along — a machine-checkable verdict nobody had put an agent inside. Recognition is that layer, and it runs today: a sentence in, a verified building out.

Honest limits: v1 designs a single storey, and the cited rulepack is Bayern residential — a workplace is designed gladly and the report says exactly what it could and could not check. A design aid, not a Prüfstatiker.

TRY IT

`0-uddeshya-0.github.io`
`/Recognition`

THE CODE

`github.com/0-uddeshya-0`
`/Recognition`

BUILT WITH

Devin · IfcOpenShell · Python
Three.js · GitHub Actions